COMP 2139 – Assignment 1

Due Date: Friday, October 16th, 2026 (11:59 pm)
Team Size: Maximum of **3 students** per group.
Project Name: **GBC_Travel-Group**-<*Number*>
Weight: 15% of your final grade.

Objective:
The goal of this assignment is to build an interactive travel booking web application, inspired by platforms such as Expedia and Travelocity, using ASP.NET Core MVC. Users of your application will be able to browse and search flights and hotels, filter and sort the results, and view the full details of any listing.

Working in your group, you will bring together front-end development, back-end development and database management, and demonstrate a solid understanding of the ASP.NET Core MVC architecture: models, views and controllers working together, backed by Entity Framework Core.

Furthermore, students should be aware that this is the first of two interlinked assignments. The skills and the foundational application developed in this assignment will be crucial for the subsequent assignment (Assignment 2), where the focus will shift to integrating more advanced features and enhancements. Some of the complex functionalities and sophisticated features are intentionally deferred to Assignment 2, allowing students to build upon their initial work and progressively enhance their application.

By the end of this project, students will have gained hands-on experience in developing a feature-rich web application that addresses real-world requirements and challenges in online travel booking. The project is designed not only to enhance technical programming skills but also to foster critical thinking, problem-solving, and collaborative work, preparing students for future endeavors in the field of web development.

Requirement:
The compulsory technologies required for this project are the following:

- **ASP.NET Core MVC**
- **C#**
- **MVC Design**
- **Entity Framework Core (with migrations)**

The **minimum** project requirements have been summarized below.

Deliverables:
Please refer to the **Assignment Submission Guidelines** at the end of this document, and to the **Demonstration Checklist & Status Report** template provided on Brightspace.

Project Requirements (Compulsory Components)

The focus for Assignment 1 is on the browsing side of the application for two travel services: **flights** and **hotels**. Users must be able to reach search from the home page, filter and sort the results, and view the details of any listing.

Car rentals, booking, data entry (create, edit and delete), and user accounts are deferred to Assignment 2, where you will build on this application.

Flight and Hotel Listings:

Requirement: Create listings for flights and hotels.

Details:

- Use seed data to populate your database with at least 5 flights and 5 hotels.
- Provide a list page for each service type showing essential information such as name, price and availability.
- Provide a details page for each listing (e.g., airline, flight times and seats available; hotel location, rating and amenities).
- Images are optional. If you include them, store an image URL or path as a string; file uploads are not required.

Search, Filtering and Sorting:

Requirement: Implement a search feature for flights and hotels.

Details:

- Provide a search form for each service type with 2 to 3 parameters (e.g., flights: origin, destination and departure date; hotels: location, minimum star rating and number of rooms needed).
- Display matching results with key details and a link to each listing's details page.
- Include at least **one filter** (e.g., price range) and **one sort option** (e.g., price low to high).
- Display a clear message when no results match the search.

Application UX (User Experience) and Design:

Requirement: Develop a user-friendly interface with clear navigation.

Details:

- A home page with direct access to flight and hotel search.
- A navbar linking to all main pages (no dead links), and a footer displaying your group number and the names of your group members.
- Use HTML, CSS (e.g., Bootstrap) and JavaScript to improve the user experience.
- See **Appendix A – Wireframes** for suggested layouts of the key pages.

Database Assignment Recommendations

The database stores the data that powers your application. Keep it focused on what Assignment 1 needs. Key considerations:

1. **Core Tables**: Your database should include tables for:
   - **Flights**: e.g., airline, flight number, origin, destination, departure and arrival times, price and seats available.
   - **Hotels**: e.g., name, location, star rating, price per night, rooms available, amenities and an optional image URL.

2. **Data Integrity**: Use data annotations for required fields, string lengths and valid ranges (e.g., price must be greater than zero, star rating from 1 to 5).

3. **Class Diagram / ERD**: Include a simple class diagram or ERD showing your entities and their properties. You will present it in your video.

4. **Migrations (compulsory)**: Use Entity Framework Core migrations to create your database. All migration files must be committed to your repository.

5. **Plan for Expansion**: Assignment 2 will add bookings, data entry and user accounts that build on these tables, so name and structure your entities cleanly and give each one a primary key.

Remember, the goal for Assignment 1 is to establish a solid foundation that not only meets the current requirements but also allows for seamless integration of more complex features in the next phase of the project. A well-designed database is crucial for the success of your application, both in its current state and as it evolves.

Final Word on Implementation

Emphasis on Creativity and Originality:

- **Wireframes as Guidance**: Appendix A provides wireframes for the key pages. They show the structure and required elements of each page, not the visual style. You are free to choose your own colours, fonts, imagery and layout details, and to add pages, as long as the required elements are present.
- **Expectation of Originality**: We encourage originality and creativity in your design choices. Remember, usability and aesthetics are subjective and will be part of the evaluation criteria.

Implementation and Problem-Solving:

- **Judgment and Implementation**: The provided requirements are detailed enough to guide you through the assignment, but they also leave room for your interpretation and decision-making in implementation.
- **Problem-Solving Skills**: Your approach to solving the challenges posed by the requirements will be a key aspect of evaluation. This is an opportunity to showcase how your group can tackle real-world problems.

Design Considerations:

- **Final Design Decisions**: All final design decisions, including the user interface, navigation, and aesthetics, are your group's responsibility.
- **MVC Focus**: Ensure that your design adheres to the MVC (Model-View-Controller) architecture, making appropriate use of models, views, and controllers. This is central to the course and critical for your project.

Flexibility and Expansion:

- **Additional Pages**: Feel free to add extra pages or features that you believe will enhance your application, as long as the core requirements are met.
- **Interpretation and Originality**: While the requirements document sets out the expected functionalities, your team has the freedom to interpret and expand upon these. Original and innovative implementations are welcome and will be considered positively in evaluations.

Assignment Submission Guidelines:

1. Video Requirement
   a. Create a Short Video presentation. Your presentation should start with an introduction, where it must display a PowerPoint (or Google Presentation), that is 1 (single) slide. The slide introduces each member of your group, again, at the very start of your video.

   b. The first (and only) slide of your presentation must include current images of you and your partner(s) (no avatars allowed) that are displayed appropriately. You must also include your Full Names, Student IDs, the Course Code, Course Name, Course Section and your Assignment information.

   c. Within the recording, you and your partner(s) will take turns demonstrating your program's functionality. You must show your site working properly. Complete the **Status Report** in the Demonstration Checklist & Status Report template provided on Brightspace, and show it during your video to confirm which requirements were successfully implemented and, for any that were not, explain why.

   d. You and your partner(s) will each share the responsibility in describing the code in your solution that drives the functionality of your program – you will want to do this part well and be very clear. Be intelligent/selective on what code segments you describe; I do not need to know how every line of code works.

   e. Sound for your video must be at an appropriate level so that your voices can be clearly heard, and your screen resolution should be set so that your program's code and console details are clearly visible. In short, QA your videos. If your video is poor, assignment failures can/will be assigned.

   f. Your video must be **no longer than 10 minutes**. There is no minimum length; a clear, focused demonstration is better than a long one. Videos that exceed 10 minutes will **not** be watched and will receive a grade of zero.

2. One team lead must submit the following components to **Brightspace** on behalf of the entire group:
   a. The completed Status Report (from the provided template) – **mandatory**
   b. The group demonstration video link (preferably a cloud recording: Zoom, Loom, Teams or YouTube) – **mandatory**
   c. Private GitHub repository URL, with your instructor given access to the repository
   d. Submit through Brightspace only. Do not submit zip files or uploaded source code, and do not email submissions.

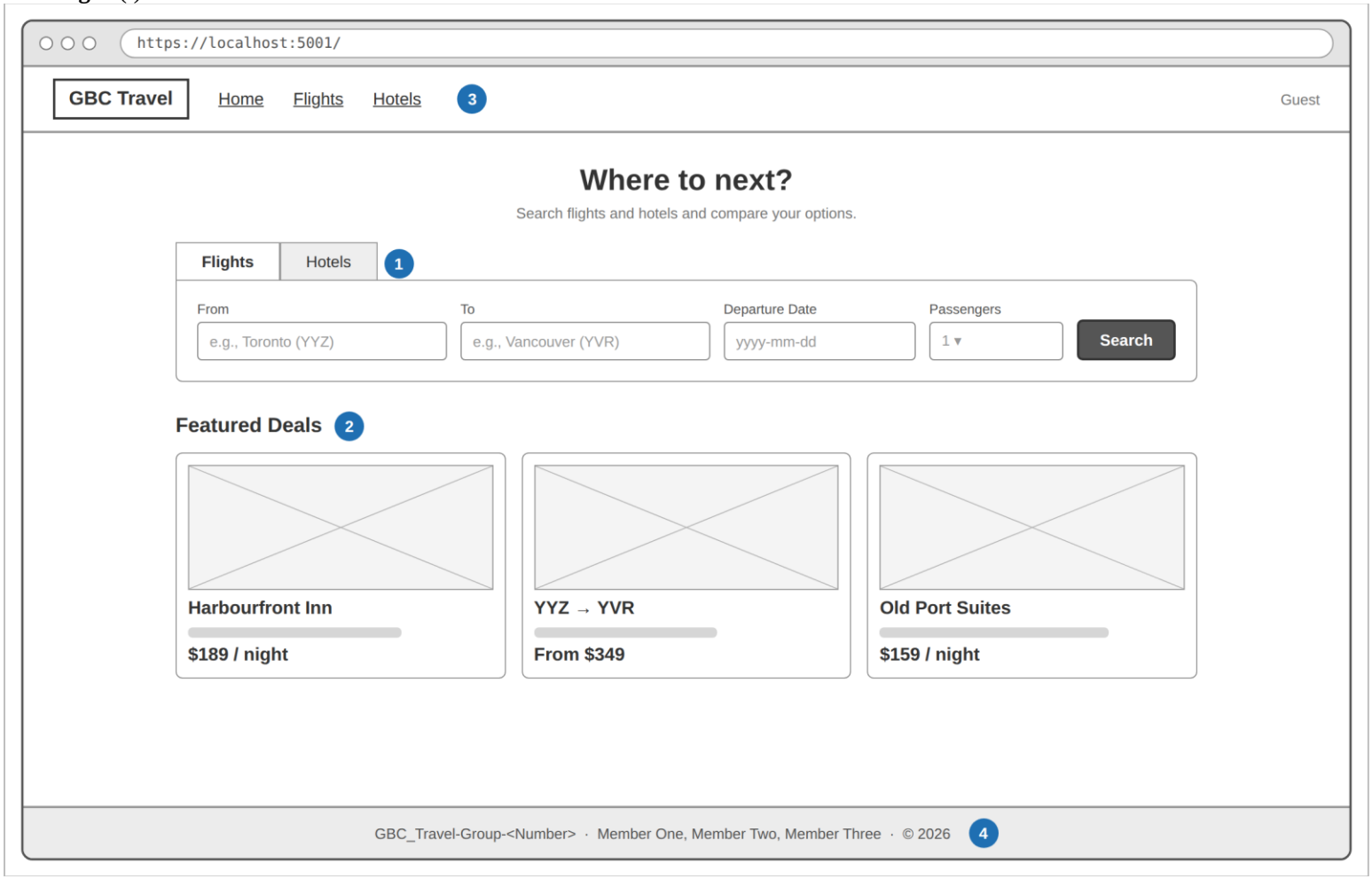
3. Be cautious **DO NOT** share your application with others. Complete failures will be assigned if code is shared. All assignments will be reviewed and analyzed strictly within these regards.

4. Late assignments are assigned a penalty of -20% per day (max 5 days).

Appendix A – Wireframes

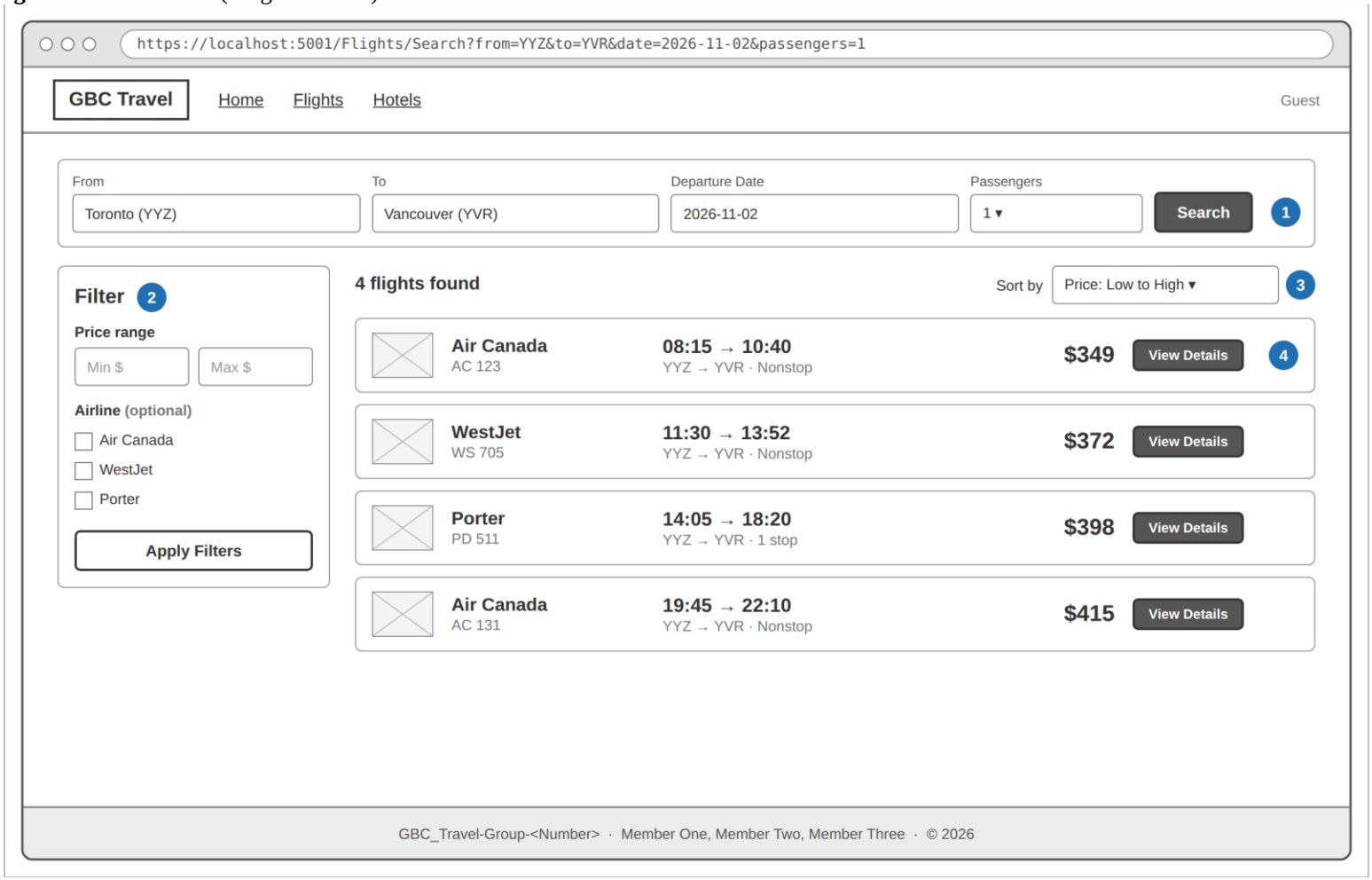
These wireframes are **guidance, not a specification**: follow the structure and include the required elements, but the visual design is your group's choice. The numbered blue markers are explained beneath each wireframe. URLs shown in the address bar are illustrative (your controller and action names may differ), and flights and hotels share the same page patterns.

Figure 1 – Home Page (/)



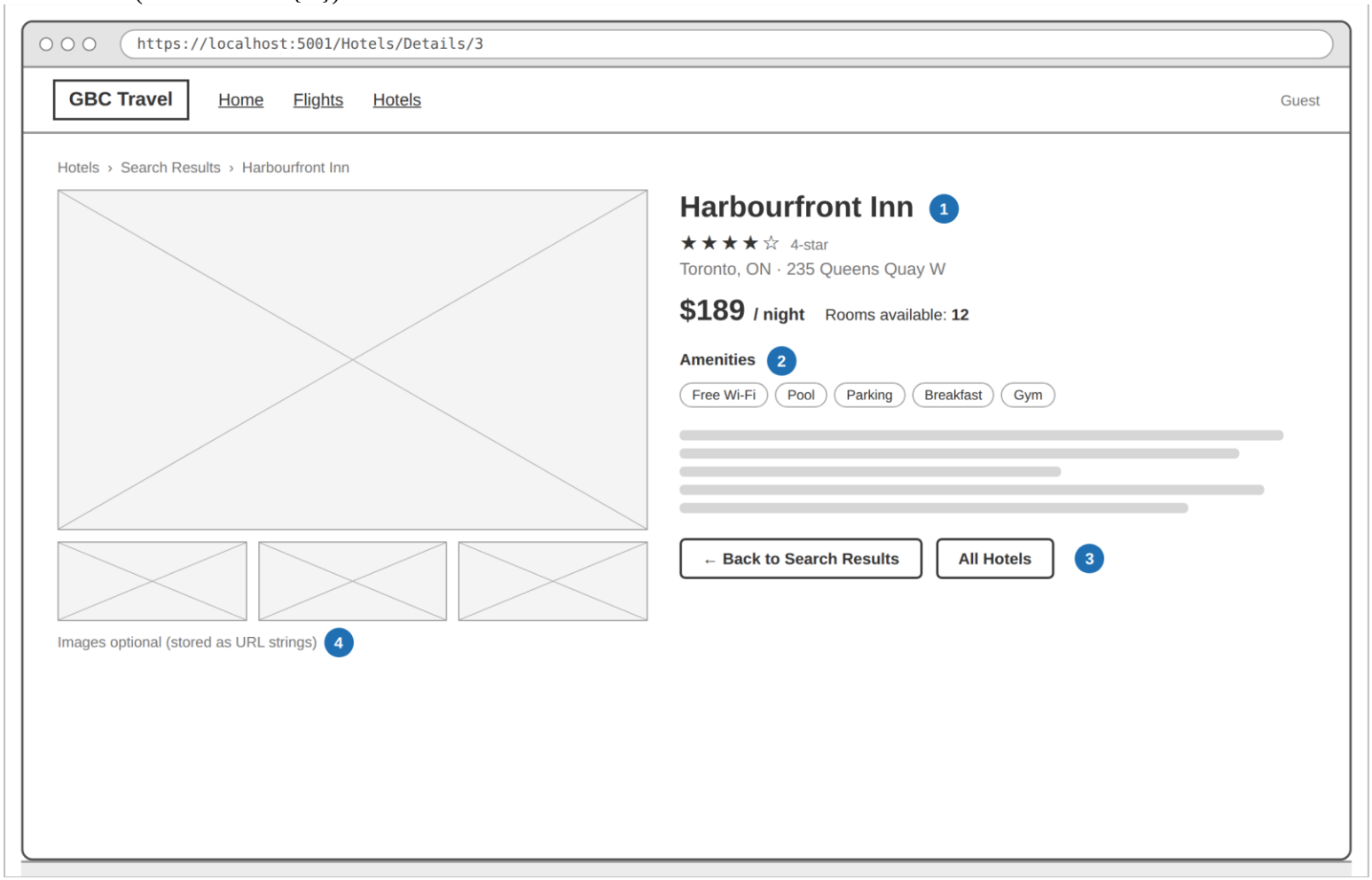
1. Search form for flights and hotels – tabs as shown, or separate search pages.
2. Featured listings (optional) – a simple way to showcase your seed data.
3. Navbar linking to all main pages (no dead links).
4. Footer displaying your group number and member names.

Figure 2 – Flight Search Results (/Flights/Search)



1 Search parameters (2 to 3). Keep the values the user entered visible on the results page.
2 At least one filter (e.g., price range). Airline checkboxes are optional.
3 At least one sort option (e.g., price low to high).
4 Each result shows key details and links to that listing's details page.
Note: When nothing matches, show a clear "no results" message in place of the list. Hotel search results follow the same pattern.

Figure 3 – Hotel Details (/Hotels/Details/{id})



1 Key details: name, rating, location, price and availability.
2 Amenities or specifications. For a flight: airline, flight number, times and seats available.
3 Navigation back to the search results or the full list of listings.
4 Images are optional; store an image URL or path as a string.
Note: Flight details pages follow the same pattern.